

Ejercicios con racionales

Multiplicar:

$$1) \quad (-2,4) \times \frac{1}{5} \times \frac{3}{2} = \left(-\frac{24}{10}\right) \times \frac{1}{5} \times \frac{3}{2} = -\frac{36}{50} = -\frac{18}{25}$$

$$2) \quad \left(-\frac{1}{2}\right) \times 2\frac{2}{3} \times (-0,3) = \left(-\frac{1}{2}\right) \times \frac{8}{3} \times \left(-\frac{3}{10}\right) = +\frac{2}{5}$$

$$3) \quad \frac{1}{4} \times (-0,1) \times \frac{1}{8} \times 1,2 = \frac{1}{4} \times \left(-\frac{1}{10}\right) \times \frac{1}{8} \times \frac{12}{10} = -\frac{3}{200}$$

$$4) \quad 0,75 \times (-4) \times \frac{1}{2} \times (-0,1) = \frac{75}{100} \times (-4) \times \frac{1}{2} \times \left(-\frac{1}{10}\right) = -\frac{75}{500} = -\frac{3}{20}$$

$$5) \quad (-2,2) \times \frac{3}{4} \times 1\frac{1}{9} \times (-0,3) = \left(-\frac{22}{10}\right) \times \frac{3}{4} \times \frac{10}{9} \times \left(-\frac{3}{10}\right) = +\frac{11}{20}$$

$$6) \quad 0,75 \times \left(-\frac{8}{9}\right) \times 3,6 \times (-1) = +\left(\frac{75}{100} \times \frac{8}{9} \times \frac{36}{10}\right) = +\frac{75 \times 8}{25 \times 10} = +\frac{3 \times 4}{5} = +\frac{12}{5}$$

Efectuar:

$$1) \quad (0,555 \dots - 1,2 \dots) \times 0,3 =$$

$$\left(\frac{5}{9} - \frac{11}{9}\right) \times \frac{3}{10} = -\frac{6}{9} \times \frac{3}{10} = -\frac{1}{5} = -0,2$$

$$2) \quad \left[1 - 0,1 \dots + \sqrt{\frac{-1}{9}}\right]^{-1} =$$

$$\left[1 - \frac{1}{9} + \sqrt{-\frac{1}{9}}\right]^{-1} = \left[1 - \frac{1}{9} + \frac{1}{3}\right]^{-1} = \left[\frac{9-1+3}{9}\right]^{-1} = \left[\frac{11}{9}\right]^{-1} = \frac{9}{11}$$

$$3) \quad 1,3 \dots - 0,066 \dots - 0,3030 \dots + \frac{2}{5} - \frac{4}{11} =$$

$$\frac{12}{9} - \frac{6}{90} - \frac{30}{99} + \frac{2}{5} - \frac{4}{11} = \frac{4}{3} - \frac{1}{15} - \frac{10}{33} + \frac{2}{5} - \frac{4}{11} = \frac{220 - 11 - 50 + 66 - 60}{165} = \frac{165}{165} = 1$$

$$4) \quad \left(\frac{3}{5} - 0,05454 \dots\right)^{-1} - 0,4 \times 0,1 \times 5 =$$

$$\left(\frac{3}{5} - \frac{54}{990}\right)^{-1} - \frac{4}{10} \times \frac{1}{10} \times 5 = \left(\frac{3}{5} - \frac{6}{110}\right)^{-1} - \frac{20}{100} = \left(\frac{3}{5} - \frac{3}{55}\right)^{-1} - \frac{1}{5} = \left(\frac{33-3}{55}\right)^{-1} - \frac{1}{5} = \left(\frac{30}{55}\right)^{-1} - \frac{1}{5} = \frac{55}{30} - \frac{1}{5} = \frac{11}{6} - \frac{1}{5} = \frac{55-6}{30} = \frac{49}{30}$$

$$5) \quad \frac{3:0,1 - \left(1 - \frac{2}{3}\right)^{-2}}{\frac{1-\frac{1}{2}}{\frac{3}{4}-\frac{1}{2}} - \frac{0,5}{\frac{1}{12}}} + \sqrt[3]{0,875 - 1} =$$

$$\frac{3:\frac{1}{10} - \left(1 - \frac{2}{3}\right)^{-2}}{\frac{1-\frac{1}{2}}{\frac{3}{4}-\frac{1}{2}} - \frac{1}{\frac{1}{12}}} + \sqrt[3]{-0,125} =$$

125 25

$$\frac{30 - \left(\frac{3-2}{3}\right)^{-2}}{\frac{1-\frac{1}{2}}{\frac{3}{4} - \frac{1}{2} - \frac{1}{2} - \frac{5}{3}}} + \sqrt[3]{-\frac{125}{1000}} =$$

$$\frac{30 - \left(\frac{1}{3}\right)^{-2}}{\frac{\frac{2-1}{2}}{\frac{3-2}{4} - \frac{3}{10}}} + \left(-\frac{5}{10}\right) =$$

$$\frac{30-9}{\frac{\frac{1}{2}-\frac{3}{10}}{\frac{1}{4}}} - \frac{1}{2} =$$

$$\frac{21}{2-\frac{3}{10}} - \frac{1}{2} =$$

$$\frac{21}{\frac{20-3}{10}} - \frac{1}{2} =$$

$$\frac{21}{\frac{17}{10}} - \frac{1}{2} =$$

$$\frac{210}{17} - \frac{1}{2} =$$

$$\frac{420-17}{34} =$$

$$\frac{403}{34} = \text{REVISAR}$$